A Tight Lower Bound for 3-Coloring Grids in the Online-LOCAL Model

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ABSTRACT

Recently, Akbari et al. (ICALP 2023) studied the locality of graph problems in distributed, sequential, dynamic, and online settings from a unified point of view. They designed a novel $O(\log n)$-locality deterministic algorithm for proper 3-coloring bipartite graphs in the Online-LOCAL model. In this work, we establish the optimality of the algorithm by showing a *tight* deterministic $\Omega(\log n)$ locality lower bound, which holds even on grids. To complement this result, we have the following additional results:

(1) We show a higher and tight $\Omega(\sqrt{n})$ lower bound for 3-coloring toroidal and cylindrical grids.
(2) Considering the generalization of 3-coloring bipartite graphs to $(k+1)$-coloring k-partite graphs, we show that the problem also has $O(\log n)$ locality when the input is a k-partite graph that admits a *locally inferable unique coloring*. This special class of k-partite graphs covers several fundamental graph classes such as k-trees and triangular grids. Moreover, for this special class of graphs, we show a tight $\Omega(\log n)$ locality lower bound.
(3) For general k-partite graphs with $k \geq 3$, we prove that the problem of $(2k-2)$-coloring k-partite graphs exhibits a locality of $\Omega(n)$ in the Online-LOCAL model, matching the round complexity of the same problem in the LOCAL model recently shown by Coiteux-Roy et al. (STOC 2024). Consequently, the problem of $(k+1)$-coloring k-partite graphs admits a locality lower bound of $\Omega(n)$ when $k \geq 3$, contrasting sharply with the $\Theta(\log n)$ locality for the case of $k = 2$.

CCS CONCEPTS

• **Theory of computation → Distributed algorithms**; **Online algorithms**.

KEYWORDS

Distributed algorithms, Online algorithms, Lower bounds, Graph coloring

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1 INTRODUCTION

We focus on the *locality* of graph problems, which has been studied in a variety of models. Throughout the paper, we only consider the *deterministic* setting.

Distributed setting: In the classical LOCAL model [17, 19] of distributed computing, an algorithm with locality T processes the nodes of a graph simultaneously in parallel, in a way that each node determines its output by examining its neighborhood with a radius of T. Intuitively, an algorithm with locality T in the LOCAL model can be run on a distributed network in T synchronous communication rounds.

Sequential setting: In the SLOCAL model [14], an algorithm with locality T processes the nodes in a sequential order that is selected by an adversary. The output of a node may depend on its T-radius neighborhood and the outputs of the previously processed nodes. For instance, the well-known *greedy coloring algorithm* solves the (degree+1)-list coloring problem with locality 1 in SLOCAL.

Dynamic setting: There have been several papers studying local algorithms in the dynamic setting [5–7, 13, 15, 16, 18], where an adversary constructs the graph *dynamically*, adding or removing nodes and edges sequentially. Following each modification, an algorithm with locality T is limited to adjusting the solution within the T-radius neighborhood of the point of change. Recently, Akbari et al. [4] formalized this setting by defining the two models Dynamic-LOCAL and Dynamic-LOCAL$^{\pm}$ to capture the incremental dynamic setting and the fully dynamic setting, respectively.

Ghaffari et al. [14] studied the connections between LOCAL and SLOCAL. They developed a method of simulating an arbitrary SLOCAL algorithm in the LOCAL model using *network decompositions*. Combining this method with the deterministic

polylogarithmic-round network decomposition of Rozhoň and Ghaffari [20], one can infer that the class of graph problems solvable with polylogarithmic locality deterministically is *identical* in LOCAL and SLOCAL.

Very recently, Akbari et al. [4] studied the locality of graph problems in distributed, sequential, and dynamic settings from a *unified* point of view. They considered a new model Online-LOCAL, which is a variant of SLOCAL that has a *global memory*. Among all the models {LOCAL, SLOCAL, Dynamic-LOCAL, Dynamic-LOCAL$^{\pm}$, Online-LOCAL}, the Online-LOCAL model is the *strongest* one: any algorithm in any of these models can be simulated in Online-LOCAL model with the same asymptotic locality. The LOCAL model is the *weakest* model in the sense that any LOCAL algorithm can be simulated in any of the above models with the same asymptotic locality. Therefore, all the mentioned models are sandwiched between the LOCAL and Online-LOCAL models. Consequently, if one can match a locality lower bound in Online-LOCAL with a locality upper bound in LOCAL, then it immediately implies a tight locality bound in all of the models.

Akbari et al. [4] obtained such a tight locality bound for a wide range of problems. For all *locally checkable labeling* problems in paths, cycles, and rooted regular trees, they have nearly the same locality in all of the models:

$$\text{LOCAL} \approx \text{SLOCAL} \approx \text{Dynamic-LOCAL}^{\pm} \approx \text{Dynamic-LOCAL} \approx \text{Online-LOCAL}.$$

Coloring bipartite graphs in Online-LOCAL. While the above result suggests that these models can be quite similar, Akbari et al. [4] demonstrated an *exponential* separation between the LOCAL and Online-LOCAL models. They designed a novel $O(\log n)$-locality algorithm for proper 3-coloring bipartite graphs in the Online-LOCAL model. In contrast, the same problem is known to have locality $\Omega(\sqrt{n})$ in the LOCAL model [8]. Their work left open the following question.

> *"Is it possible to find a 3-coloring in bipartite graphs in the* Online-LOCAL *model with locality $o(\log n)$?"*

In this work, we resolve the question by demonstrating a *tight* $\Omega(\log n)$ locality lower bound in the Online-LOCAL which holds even on grids, which are bipartite graphs.

Theorem 1.1. *In the* Online-LOCAL *model, the locality of 3-coloring a $(\sqrt{n} \times \sqrt{n})$ grid is $\Omega(\log n)$.*

Combining the above result with the $O(\log n)$ upper bound by Akbari et al. [4], a tight bound is obtained.

Corollary 1.2. *In the* Online-LOCAL *model, the locality of 3-coloring bipartite graphs is $\Theta(\log n)$.*

To establish Theorem 1.1, a main technical barrier that we overcome is that in Online-LOCAL, we cannot rely on an *indistinguishability* argument due to the presence of global memory. Most of the existing LOCAL lower bound proofs rely on such kind of an argument. In particular, the known $\Omega(\sqrt{n})$ lower bound for 3-coloring grids in the LOCAL model [8] works by first proving the lower bound on *toroidal* grids and then using an indistinguishability argument to extend the lower bound to grids. The lower bound argument in [8] heavily depends on the assumption that the underlying graph is a toroidal grid with an *odd* number of columns, which is *not* bipartite.

The presence of a global memory in Online-LOCAL allows the possibility for a graph problem to have different localities in grids and toroidal grids. Indeed, in this work, we show a much higher $\Omega(\sqrt{n})$ lower bound for 3-coloring on toroidal grids in the Online-LOCAL model. The lower bound does not contradict the $O(\log n)$-locality algorithm of [4] because toroidal grids are not bipartite in general.

Theorem 1.3. *In the* Online-LOCAL *model, the locality of 3-coloring $(\sqrt{n} \times \sqrt{n})$ toroidal and cylindrical grids is $\Omega(\sqrt{n})$.*

Combining the above Online-LOCAL lower bound with the trivial $O(\sqrt{n})$ locality upper bound in the LOCAL model, we establish that $\Theta(\sqrt{n})$ is a *tight* locality bound for 3-coloring toroidal and cylindrical grids in all the models discussed above, as they are sandwiched between LOCAL and Online-LOCAL.

Corollary 1.4. *The locality of 3-coloring $(\sqrt{n} \times \sqrt{n})$ toroidal and cylindrical grids is $\Theta(\sqrt{n})$ in all the models* {LOCAL, SLOCAL, Dynamic-LOCAL, Dynamic-LOCAL$^{\pm}$, Online-LOCAL}.

Interestingly, our results yield two alternative proofs for the $\Omega(\sqrt{n})$ locality lower bound for 3-coloring grids in LOCAL [8], which we briefly explain as follows. By an indistinguishability argument, the $\Omega(\sqrt{n})$ locality lower bound in Corollary 1.4 on toroidal and cylindrical grids in LOCAL automatically implies the same lower bound on grids in LOCAL. Alternatively, the $\Omega(\log n)$ lower bound of Theorem 1.1 in Online-LOCAL implies the same lower bound in LOCAL, and we can automatically improve such a lower bound to $\Omega(\sqrt{n})$ by the known $\omega(\log^* n)$–$o(\sqrt{n})$ complexity gap on grids in LOCAL [10, 11].

Generalization: $(k+1)$-coloring k-partite graphs in Online-LOCAL. Next, we consider a natural generalization of the 3-coloring problem in bipartite graphs to the task of $(k+1)$-coloring of k-partite graphs for any constant $k \geq 2$ within the Online-LOCAL model. Given Corollary 1.2, a pertinent question arises:

- Is the locality of $(k+1)$-coloring k-partite graphs in Online-LOCAL $\Theta(\log n)$ for all k?

Interestingly, for $k \geq 3$, we can establish a lower bound of $\Omega(n)$. More specifically, we present a *much stronger* result demonstrating an $\Omega(n)$ lower bound for $(2k-2)$-coloring.

Theorem 1.5. *Let $k \geq 2$ be a constant. The locality of $(2k-2)$-coloring a k-partite graph is $\Omega(n)$ in the* Online-LOCAL *model.*

Observe that $k+1 \leq 2k-2$ for $k \geq 3$, so we have the following corollary.

Corollary 1.6. *Let $k \geq 3$ be a constant. The locality of $(k+1)$-coloring a k-partite graph is $\Omega(n)$ in the* Online-LOCAL *model.*

Recall that for the case of $k = 2$, the locality $(k+1)$-coloring a k-partite graph is already known to be $\Theta(\log n)$ by Corollary 1.2.

Related work. In the LOCAL model, the same lower bound of $\Omega(n)$ for $(2k-2)$-coloring k-partite graphs has been very recently established in [12]. More generally, they showed that the round complexity of the problem of c-coloring k-partite graphs is $\widetilde{\Theta}(n^{1/\alpha})$ in LOCAL, where $\alpha = \left\lfloor \frac{c-1}{k-1} \right\rfloor$ and the notation $\widetilde{\Theta}(\cdot)$ hides polylogarithmic factors. Notably, this bound is polynomial in n for any constants k and c.

For the case of $(c,k) = (3,2)$, there is a huge gap between the complexities of 3-coloring bipartite graphs in LOCAL ($\widetilde{\Theta}(\sqrt{n})$, [12]) and Online-LOCAL ($\Theta(\log n)$, Corollary 1.2). On the other hand, for $k \geq 3$ and $c \leq 2k-2$, Theorem 1.5 says that the locality of c-coloring k-partite graphs is $\Omega(n)$ in Online-LOCAL – a match with the round complexity of the same problem in LOCAL.

Improved upper bounds for special graph classes. For the problem of $(k+1)$-coloring of k-partite graphs in Online-LOCAL, the locality is $\Theta(\log n)$ for $k = 2$ (Corollary 1.2) and is $\Omega(n)$ for $k \geq 3$ (Corollary 1.6). A plausible explanation for this notable shift in the locality bound when transitioning from $k = 2$ to $k \geq 3$ is as follows: While the bipartition (i.e., 2-coloring) of a (connected) bipartite graph is unique, a k-partite graph might not admit a unique k-coloring in general. Intuitively, this means that for $k \geq 3$, the colorings of two graph fragments computed by an Online-LOCAL algorithm may be completely incompatible, requiring more adjustments to combine them, thus leading to a locality of $\Omega(n)$.

We show that it is possible to break the $\Omega(n)$ locality lower bound of Theorem 1.5 if the underlying k-partite graph admits a unique k-coloring satisfying the following desirable property: For each graph fragment seen by an Online-LOCAL algorithm, the unique k-coloring restricted to that fragment can be inferred correctly, up to a permutation of the k colors. As we will later see, examples of such graphs include all (connected) *bipartite graphs*, *triangular grids*, and *k-trees*.

Let $\ell \in \mathbb{N}$ be a non-negative integer. Let G be a k-partite graph. Let $G' = (V', E')$ be a connected subgraph of G. Define $\mathcal{C}^*(G', \ell)$ as the set of all proper k-colorings of the subgraph of G induced by the ℓ-radius neighborhood $\mathcal{B}(V', \ell)$ of V'. Define $\mathcal{C}(G', \ell)$ as the set of all proper k-colorings of G' resulting from restricting some coloring $c \in \mathcal{C}^*(G', \ell)$ to the domain V'. We say that two k-colorings c and c' are identical *up to a permutation* if there exists a permutation $\phi\colon [k] \to [k]$ such that $c' = \phi \circ c$.

Definition 1.7 (Locally inferable unique colorings). We say that a k-partite graph G has a locally inferable unique k-coloring with radius ℓ if all colorings in $\mathcal{C}(G', \ell)$ are identical up to a permutation for any connected subgraph G' of G.

To put it another way, G admits a locally inferable unique k-coloring with radius ℓ if the k-color partition of G is unique and can be inferred locally in the following way: For any connected subgraph G' of G, the unique k-color partition restricted to G' can be inferred by inspecting its ℓ-radius neighborhood. Since the bipartition of a bipartite graph is unique, every bipartite graph has a locally inferable unique k-coloring with radius $\ell = 0$. We show that triangular grids and k-trees also admit locally inferable unique colorings with $\ell \in O(1)$.

Triangular grids: The triangular grid $G = (V, E)$ of side length d is defined as follows.

- $V = \{(x, y) \in \mathbb{N}^2 : 0 \leq x + y \leq d\}$.
- For any two nodes (x, y) and (x', y') in V, add an edge between them to E if $|x - x'| + |y - y'| = 1$ or $x - x' = y - y' \in \{-1, 1\}$.

Consider a connected subgraph $G' = (V', E')$ of a triangular grid G. See Figure 1 for an illustration. For any node $u \in V'$, there is a triangle Δ_u containing u in the 1-radius neighborhood of V' in G. Observe that for any two nodes u and v in V', they are connected by a sequence of triangles $(\Delta_u = \Delta_1, \Delta_2, \ldots, \Delta_t = \Delta_v)$ in the 1-radius neighborhood of V' in G: For each $i \in [t-1]$, Δ_i and Δ_{i+1} shares an edge. If the nodes of Δ_i and Δ_{i+1} are $\{a, b, c\}$ and $\{b, c, d\}$, respectively, then the color of a and d must be identical under any 3-coloring. In particular, fixing the coloring of Δ_u uniquely determines the color of v. Therefore, for any given 3-coloring of the 1-radius neighborhood of V' in G, its restriction to V' must be unique up to a permutation. In other words, G admits a locally inferable unique 3-coloring with radius $\ell = 1$.

***k*-trees:** The graphs that can be constructed as follows are called k-trees: Start with a $(k+1)$-clique, and then iteratively add nodes such that each newly added node v is adjacent to exactly k existing nodes that form a k-clique. The construction of a k-tree G naturally yields a tree H so that each node $h \in V(H)$ corresponds to a $(k+1)$-clique of G and two nodes of H are adjacent if their corresponding cliques share k nodes. Consider a connected subgraph $G' = (V', E')$ of a k-tree G. Similar to the case of triangular grids, by querying the 1-radius neighborhood of V' in G, we learn all the $(k+1)$-cliques containing at least one node in V' and their adjacency relations in H. Therefore, k-trees admit a locally inferable unique $(k+1)$-coloring with radius $\ell = 1$, as fixing the color assignment of any one of these $(k+1)$-cliques fixes the color of all other nodes in G'.

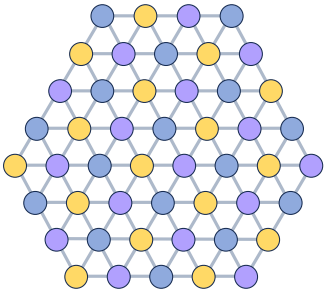


Figure 1: The unique tripartition of a triangular grid restricted to a connected subgraph.

For notational simplicity, from now on, we write $\mathcal{L}_{k,\ell}$ to denote the set of all k-partite graphs having a locally inferable unique k-coloring with radius ℓ. We show that a $(k+1)$-coloring for any graph in $\mathcal{L}_{k,\ell}$ with $k \in O(1)$ and $\ell \in O(1)$ can be computed with $O(\log n)$ locality in the Online-LOCAL model.

Theorem 1.8. *Let $k \geq 2$ be any constant. There is an* Online-LOCAL *algorithm with locality $O(\log n)$ that $(k+1)$-colors any graph in $\mathcal{L}_{k,\ell}$ with $\ell \in O(1)$.*

The following corollary is an immediate consequence of Theorem 1.8.

Corollary 1.9. *The locality of 3-coloring triangular grids and $(k+2)$-coloring k-trees with $k \in O(1)$ is $O(\log n)$ in the* Online-LOCAL *model.*

We complement Theorem 1.8 with a matching lower bound by extending the lower bound of Theorem 1.1 from $k = 2$ to higher values of k.

Theorem 1.10. *Let $k \geq 2$ be any constant. For any* Online-LOCAL *algorithm that $(k+1)$-colors any graph in $\mathcal{L}_{k,\ell}$ with $\ell \in O(1)$, its locality must be $\Omega(\log n)$.*

Combining Theorems 1.8 and 1.10, we obtain a tight bound for $(k+1)$-coloring for $\mathcal{L}_{k,\ell}$ with $\ell \in O(1)$.

Corollary 1.11. *For any constant $k \geq 2$, in the* Online-LOCAL *model, the locality of $(k+1)$-coloring graphs in $\mathcal{L}_{k,\ell}$ with $\ell \in O(1)$ is $\Theta(\log n)$.*

Related models. Subsequent to the initial publication of this paper, Akbari et al. [3] studied the *randomized* variant of the Online-LOCAL model, extending our $\Omega(\log n)$ lower bound to the randomized setting. In addition to the Online-LOCAL model, researchers have studied other hybrids of shared-memory and message-passing models, including the *message-and-memory* (m&m) model [2] and the *computing-with-the-cloud* (CWC) model [1].

1.1 Technical Overview

We give an overview of the proofs of Theorems 1.1, 1.3, 1.5, 1.8 and 1.10.

1.1.1 Lower Bound for 3-Coloring Bipartite Graphs. We start by reviewing the $\Omega(\sqrt{n})$ locality lower bound for 3-coloring grids in LOCAL [8]. First of all, by an indistinguishability argument, they can focus on toroidal grids and not grids. Without loss of generality, they assumed that the coloring has a special property that any node colored with 3 is adjacent to a node colored with 1 and a node colored with 2. They showed that for any proper 3-coloring of a toroidal grid satisfying the above condition, it is possible to orient the diagonals between any two nodes colored with 3 in such a way that yields an Eulerian graph over the set of all nodes colored with 3.

They made a key observation that the difference between the number of times a directed edge in the Eulerian graph passes a row in the upward direction and in the downward direction is *invariant* of the choice of the row. Let $s(\ell)$ denote this value, where ℓ is the number of nodes in a row. They showed that when ℓ is odd, $s(\ell)$ must be an odd number within the range $[-\ell/2, \ell/2]$. Based on this property, they obtained the desired lower bound by a reduction to the following problem on an n-cycle, which requires $\Omega(n)$ rounds to solve: Each node outputs a number in such a way that all the numbers sum up to exactly q, where q is any given odd integer in the range $[-n/2, n/2]$.

The above lower bound proof does *not* apply to Online-LOCAL because it highly depends on the assumption that the underlying graph is a toroidal grid, which is fine in the LOCAL model due to an indistinguishability argument. This is not possible for Online-LOCAL due to the presence of a global memory. To extend the lower bound to Online-LOCAL, new ideas are required.

Our Online-LOCAL lower bound starts from the following informal observation. The notion of $s(\ell)$ can be extended beyond rows. Informally, for any directed path or directed cycle, we may define its s-value as the difference between the number of times a directed edge in the Eulerian graph passes it in the rightward direction and the leftward direction. Similarly, it is possible to argue that if two directed paths or two directed cycles are *homotopic* to each other, then their s-values must be identical. Moreover, this observation still makes sense even if the underlying graph is a grid and not a toroidal grid, although the directed graph defined on the set of all nodes colored with 3 might not be Eulerian in that case.

For the case of grids, one can observe that the s-value of any directed cycle must be zero. This gives rise to the following proof idea in the Online-LOCAL model. Suppose we have an adversary strategy that can force an algorithm to create a directed path in a row on the grid that has an s-value of at least t. In that case, such an algorithm must have locality $\Omega(t)$, since otherwise using such a directed path construction, the adversary can force a cycle to have an s-value that is nonzero, which is a contradiction.

A technical difficulty in realizing the above proof idea is that in the informal definition of the s-value of a directed path or directed cycle suggested above, the s-value not only depends on the colors on the path or cycle under consideration but also depends on the colors of all nodes adjacent to the path or cycle. Moreover, there are subtle issues about how we deal with an intersection that occurs at the endpoints of a path. This makes the informal definition of s-value inconvenient to use, especially because we have to consider partial coloring to prove lower bounds in the Online-LOCAL model.

To deal with the above issue, we develop the notion of b-value that depends *only* on the coloring of the considered path or cycle. The notion of b-value can be seen as a simplification of the notion of s-value that still enjoys all the nice properties of s-value that we need. It is noteworthy that our notion of b-value does not require the assumption that any node colored with 3 is adjacent to a node colored with 1 and a node colored with 2.

To realize the above proof idea, we demonstrate a recursive strategy that constructs a directed path with length at most $T \cdot c^k$ whose b-value is at least k, for some constant $c > 0$ and any given T-locality Online-LOCAL algorithm. By setting both T and k to be $\Theta(\log n)$ with a sufficiently small leading constant, the path length can be made smaller than $\sqrt{n}$, so we can fit the path in a $(\sqrt{n} \times \sqrt{n})$ grid. This gives us the desired $\Omega(\log n)$ locality lower bound. A key idea in the recursive construction is to utilize a property of b-value that whether it is even or odd is determined by the parity of the path length and the colors of the two endpoints.

To show the higher lower bound of $\Omega(\sqrt{n})$ in cylindrical and toroidal grids, we again utilize the property that the b-value of each row is invariant of the choice of the row and is odd given that each row contains an odd number of nodes. Therefore, as an adversary, we may just ask the algorithm to produce a coloring of two rows, and then we set their directions in a way to force them to have different b-values.

1.1.2 Lower Bound for $(2k-2)$-Coloring k-Partite Graphs. To establish the $\Omega(n)$ lower bound for $(2k-2)$-coloring k-partite graphs in the LOCAL model, [12] considered an instance $G^{\#}$ such that the chromatic number of $G^{\#}$ is $2k-1$, while the subgraphs induced by the $o(n)$-radius neighborhood of any node of $G^{\#}$ are k-partite. It is crucial to note that $G^{\#}$ lies outside the input family. Despite being outside the input family, the instance is locally solvable, signifying that the subgraph induced by the $o(n)$-neighborhood of any node can be colored with k colors. The lower bound in the LOCAL model is established through an indistinguishability argument by Linial [17], which proves lower bounds by employing hard instances taken from outside the input family: To show a T-round LOCAL lower bound for c-coloring for G by this approach, one just needs to construct a graph $G^{\#}$ that cannot be c-colored and is indistinguishable to G by a T-round LOCAL algorithm.

The construction of the hard instance k-partite graph G^*, to establish the lower bound for $(2k-2)$-coloring k-partite graphs in the Online-LOCAL model, is inspired by the construction of $G^{\#}$. However, the application of Linial's technique, designed for LOCAL, is not straightforward in the Online-LOCAL model possibly due to the presence of global memory. Despite the inspiration of our hard instance being inspired from the hard instance in the LOCAL model, our arguments to establish the lower bound in the Online-LOCAL model are different, as sketched in the subsequent paragraphs, and accommodate the unique features of the Online-LOCAL setting, including the utilization of global memory.

In the construction of the hard instance graph G^*, we consider n/k^2 gadgets, each comprising k^2 nodes. The node set of a gadget is represented as elements in a $k \times k$ matrix. Two nodes in a gadget are connected by an edge if and only if they are neither in the same row nor the same column. Under a proper coloring of a gadget, the gadget is said to be *row-colorful* (*column-colorful*) if there exists a row (column) with all the k nodes being colored by distinct colors. We show that a gadget is exactly one out of row-colorful and column-colorful when we have a proper coloring of the gadget with $2k-2$ colors. G^* is formed by assembling n/k^2 gadgets, connecting nodes in consecutive gadgets as follows: for any two consecutive gadgets, two nodes (one from each) are connected by an edge if and only if they are neither in the same row nor the same column. We prove that, in any proper coloring of G^* with $2k-2$ colors, either all the gadgets are row-colorful or all of them are column-colorful.

The basis of our lower bound argument is as follows: if the locality of the algorithm in the Online-LOCAL model is $o(n)$, then the adversary can ask the algorithm to color G^* in a specific order such that the first gadget is row-colorful and the last gadget is column-colorful. This scenario results in the impossibility of properly coloring some nodes, as all the gadgets must be either row-colorful or column-colorful.

In [12], the authors consider the problem of c-coloring k-partite graphs, where c and k are constants. This is a more general version than both 3-coloring bipartite graphs and $(2k-2)$-coloring k-partite graphs. Specifically, [12] establishes that the round complexity of this generalized problem is essentially $\widetilde{\Theta}(n^{1/\alpha})$, where $\alpha = \lfloor \frac{c-1}{k-1} \rfloor$. We have discussed the extension of the hard instance construction for $c = 2k-2$ (and thus $c \leq 2k-2$) to demonstrate an $\Omega(n)$ lower bound for $(2k-2)$-coloring k-partite graphs in the Online-LOCAL model. It is natural to consider the possibility of such an extension for all c and k. However, if such a general extension is possible for all c and k, we would have achieved $\Omega(\sqrt{n})$ lower bound on the locality of 3-coloring bipartite graphs in the Online-LOCAL model, which is impossible as the locality of 3-coloring bipartite graphs is $\Theta(\log n)$ due to Corollary 1.2. Resolving the locality of c-coloring k-partite graphs for all values of c and k in the Online-LOCAL model remains an intriguing open problem.

1.1.3 Coloring Graphs with Locally Inferable Unique Colorings. From the upper bound side, Akbari et al. [4] presented an $O(\log n)$-locality Online-LOCAL algorithm of 3-coloring a bipartite graph. Their algorithm is based on a key observation that the bipartition for any (connected) bipartite graph G is unique, so there are exactly two different ways to 2-color G. In the subsequent discussion, we refer to these two possibilities as two *parities*.

The algorithm of Akbari et al. [4], in a nutshell, tries to 2-color the graph fragments seen by the Online-LOCAL algorithm using the colors in $\{1, 2\}$. A technical challenge arises when two graph fragments with different parities become connected, as their 2-coloring are incompatible with each other. The issue is resolved by *flipping* the parity of one graph fragment by adding a layer of the third color on the boundary. The flipping operation requires $O(1)$ locality. By always flipping the graph fragment with a smaller size, it is guaranteed that at most $O(\log n)$ flipping operations are needed from the perspective of each node. Therefore, the total required locality is $O(\log n)$.

In this work, we extend their algorithm to $(k+1)$ coloring all k-partite graphs having a locally inferable unique coloring with radius $\ell \in O(1)$. For these k-partite graphs, there is a unique partition into k parts that can be inferred locally: Given a connected node set S, the partition restricted to S can be inferred by information within the ℓ-radius neighborhood of S. Therefore, we can assume that there is an oracle $\mathcal{O}$ that returns such a k-partition for any graph fragment seen by the Online-LOCAL algorithm, as the cost of implementing such an oracle is just an extra $\ell \in O(1)$ locality. Our $(k+1)$ coloring algorithm is based on the generalization of the notion of parity into the assignment of the *indices* $\{1, 2, \ldots, k\}$ to the k parts in the partition, which is possible due to the oracle $\mathcal{O}$. Similar to the algorithm of Akbari et al. [4], we try to k-color the graph fragments seen by the Online-LOCAL algorithm using the colors in $[k]$. When two graph fragments with incompatible k-coloring become connected, we show that with $O(1)$ locality, it is possible to unify their coloring by iteratively swapping the indices with the help of the extra color $k+1$. As such, we can $(k+1)$-color any graph in $\mathcal{L}_{k,\ell}$ for any constants k and ℓ with $O(\log n)$ locality.

Now, we discuss the idea behind $\Omega(\log n)$ lower bound on the locality of $(k+1)$-coloring k-partite graphs when the input graph is from $\mathcal{L}_{k,\ell}$ with $\ell \in O(1)$. Recall that Theorem 1.1 already establishes this result for the case of $k = 2$, as grids are bipartite graphs, which belong to $\mathcal{L}_{2,0}$. To extend the result to higher values of k, we construct a sequence of graphs $(G_2, G_3, \ldots, G_k)$ recursively such that $G_k \in \mathcal{L}_{k,\ell}$ with $\ell \in O(1)$. For the base case, G_2 is a $(\sqrt{n} \times \sqrt{n})$ grid. For the inductive step, G_{i+1} is constructed from G_i by augmenting each node u in G_i with a new node u' that is connected to u and its (original) neighbors. We show that an $o(\log n)$-locality algorithm that properly colors G_i with $i+1$ colors can be obtained

from an $o(\log n)$-locality algorithm that properly colors G_{i+1} with $i+2$ colors, so the $\Omega(\log n)$ lower bound for the case of $k=2$ (Theorem 1.1) implies the same asymptotic lower bound for any constant $k \geq 2$.

1.2 Roadmap

In Section 2, we introduce essential graph terminology and define the computation models. In Section 3, we present our main lower bound results: an $\Omega(\log n)$ locality lower bound for 3-coloring in simple grids and a higher $\Omega(\sqrt{n})$ lower bound for toroidal and cylindrical grids (Theorems 1.1 and 1.3). In Section 4, we prove the $\Omega(n)$ locality lower bound for $(2k-2)$-coloring of k-partite graphs (Theorem 1.5). In the full version [9] of the paper, we establish a tight $\Theta(\log n)$ bound for $(k+1)$-coloring in locally inferable unique coloring graphs (Theorems 1.8 and 1.10).

2 PRELIMINARIES

Throughout this paper, the input graph $G=(V,E)$ is simple, connected, undirected, and finite unless explicitly specified otherwise. We write $n=|V|$ to denote the number of nodes in G. For a subset of nodes $U \subseteq V$ and a natural number T, we write $\mathcal{B}(U,T)$ to denote the set of all nodes within the T-radius neighborhood of any $v \in U$. If there is only one node v in U, we may write $\mathcal{B}(v,T)$ instead of $\mathcal{B}(U,T)$. We write $G[U]$ to denote the subgraph of G induced by the set of nodes $U \subseteq V$. In particular, $G[\mathcal{B}(v,T)]$ is the subgraph of G induced by the T-radius neighborhood of v. For any graph H, we write $V(H)$ and $E(H)$ to denote its node set and edge set, respectively. For a subgraph $H \subseteq G$ and a node u, we write $H-u$ to denote the induced subgraph $H[V(H) \setminus \{u\}]$. For $x \in \mathbb{N}$, we write $[x]$ to denote $\{1,2,\ldots,x\}$.

2.1 Grids

We investigate local algorithms in three types of grid topologies.

- **Simple grids:** A simple grid of dimension $(a \times b)$ is a grid with a rows and b columns. More formally, an $(a \times b)$ grid is defined by the set of nodes $\{(i,j) : i \in [a] \text{ and } j \in [b]\}$ such that two nodes (i,j) and (i',j') are adjacent if and only if $|i-i'|+|j-j'|=1$. For each $i \in [a]$, the set of nodes $\{(i,j) : j \in [b]\}$ forms a *row*. For each $j \in [b]$, the set of nodes $\{(i,j) : i \in [a]\}$ forms a *column*. In a simple grid, each row and each column induce a path.
- **Cylindrical grids:** A cylindrical grid of dimension $(a \times b)$ is the result of adding the edges in $\{\{(i,1),(i,b)\} : i \in [a]\}$ to an $(a \times b)$ grid. In other words, a cylindrical grid is the result of connecting the left border and the right border of a simple grid. In a cylindrical grid, each row induces a cycle and each column induces a path.
- **Toroidal grids:** A toroidal grid of dimension $(a \times b)$ is the result of adding the edges in $\{\{(i,1),(i,b)\} : i \in [a]\}$ and $\{\{(1,j),(a,j)\} : j \in [b]\}$ to an $(a \times b)$ grid. In other words, a toroidal grid is the result of connecting the left and right borders and connecting the upper and lower borders of a simple grid. In a toroidal grid, each row and each column induce a cycle.

In Theorem 1.1, we consider simple grids of dimension $\left(\sqrt{n} \times \sqrt{n}\right)$. In Theorem 1.3, we consider toroidal and cylindrical grids of dimension $\left(\sqrt{n} \times \sqrt{n}\right)$. It is worth noting that toroidal grids and cylindrical grids with an odd number of columns are not bipartite graphs.

2.2 Models

We proceed to define two computation models: LOCAL and Online-LOCAL. In these models, the complexity of an algorithm is measured by its *locality*, which informally refers to the minimum number T such that the output of a node is determined by its T-radius neighborhood. Generally, the locality of an algorithm can be a function of the number $n=|V|$ of nodes in the input graph $G=(V,E)$. It is assumed that the algorithm knows the value of n. Throughout the paper, we only consider *deterministic* algorithms, so it does not matter whether the adversary is oblivious or adaptive in the following definitions. In both LOCAL and Online-LOCAL, the adversary assigns unique identifiers from the set $\{1,2,\ldots,\text{poly}(n)\}$ to the nodes.

The LOCAL *model.* In an algorithm with locality T in the LOCAL model [17, 19], each node independently determines its part of the output based on the information within its T-radius neighborhood. For example, for the proper k-coloring problem, each node $v \in V$ just needs to output its own color $c(v) \in [k]$ in such a way that no two adjacent nodes output the same color. The output of a node may depend on factors such as the graph structure, input labels, and unique identifiers.

There is an alternative way of defining the LOCAL model from the perspective of distributed computing by viewing the input graph $G=(V,E)$ as a communication network, where each node $v \in V$ is a machine and each edge $e \in E$ is a communication link. The communication proceeds in synchronous rounds. In each round, each node can communicate with its neighbors by exchanging messages of unlimited size. The locality of an algorithm is the number of communication rounds.

The Online-LOCAL *model.* In an algorithm of locality T in the Online-LOCAL model [4], nodes are processed sequentially according to an adversarial input sequence $\sigma=(v_1,v_2,\ldots,v_n)$. Let $G_i = G\left[\bigcup_{j=1}^{i} \mathcal{B}(v_j,T)\right]$ denote the subgraph induced by the T-radius neighborhoods of the first i nodes in σ. When the adversary presents a node v_i, the algorithm must assign an output label to v_i based on the information from the subsequence $(v_1,v_2,\ldots,v_i)$ and the induced subgraph G_i.

In the related SLOCAL model [14], the assignment of the output label of v_i can *only* be based on $G[\mathcal{B}(v_i,T)]$ and the output labels that were already assigned to the nodes in $\mathcal{B}(v_i,T)$, so Online-LOCAL can be seen as a variant of SLOCAL that has a global memory.

3 HARDNESS OF 3-COLORING IN GRIDS

In this section, we establish our lower bound results for 3-coloring in a simple grid, as well as toroidal and cylindrical grids. In Section 3.1, we discuss some groundwork to prove the lower bounds presented in the subsequent sections. Moving on to Section 3.2, we prove Theorem 1.1, showing the $\Omega(\log n)$ locality lower bound for 3-coloring in a simple grid. Subsequently, in Section 3.3, we prove

Theorem 1.3, establishing the $\Omega(\sqrt{n})$ locality lower bound for 3-coloring in toroidal and cylindrical grids.

3.1 Properties of 3-Coloring in Grids

In this section, we focus on analyzing the properties of proper 3-coloring and introducing the notion of b-value, which plays a critical role in the proof of Theorems 1.1 and 1.3. Although some of the results in this section apply to general 3-partite graphs, they are particularly interesting when we restrict our focus to grid topologies. Throughout this section, we let $c\colon V \to \{1,2,3\}$ be any proper 3-coloring of any 3-partite graph $G = (V, E)$.

Definition 3.1 (a-value). Let $\{u, v\} \in E$ be an edge in G. Define the a-value of (u, v) by

$$a(u,v) = \begin{cases} c(u) - c(v) & \text{if } c(u) \neq 3 \text{ and } c(v) \neq 3, \\ 0 & \text{otherwise.} \end{cases}$$

For any edge $\{u, v\} \in E$, note that $a(u, v) \in \{-1, 0, 1\}$. Moreover, the a-value of (u, v) is non-zero if and only if one of $\{u, v\}$ is colored with 1 and the other one is colored with 2. Observe that we always have $a(u, v) + a(v, u) = 0$ for any edge $\{u, v\} \in E$.

We define the b-value of a directed path P or a directed cycle C by the sum of $a(u, v)$ over all directed edges (u, v) in P or C, respectively.

Definition 3.2 (b-value). Let H be a directed path or a directed cycle. Define the b-value of H by

$$b(H) = \sum_{(u,v)\in H} a(u,v).$$

For the special case of a zero-length directed path P, we define $b(P) = 0$, as it does not contain any directed edge.

We observe that $b(C) = 0$ for any 4-node directed cycle C. This observation is particularly relevant to grid topologies, as they consist of "cells" that are 4-node cycles.

Lemma 3.3 (Cancellation of the a-values within a cell). *For any 4-node directed cycle C, $b(C) = 0$.*

Proof. Let $C = \begin{matrix} t & \leftarrow & s \\ \downarrow & & \uparrow \\ u & \rightarrow & v \end{matrix}$, so we have $b(C) = a(u, v) + a(v, s) + a(s, t) + a(t, u)$. Since there are only three colors, we have $c(u) = c(s)$ or $c(v) = c(t)$ or both. Without loss of generality, we assume $c(u) = c(s)$, so

$$a(u,v) + a(v,s) = a(u,v) + a(v,u) = 0$$

and

$$a(s,t) + a(t,u) = a(s,t) + a(t,s) = 0,$$

implying that $b(C) = a(u, v) + a(v, s) + a(s, t) + a(t, u) = 0$. □

We say that a cycle is *simple* if it does not have repeated nodes. As a consequence of Lemma 3.3, we show that $b(C) = 0$ for any simple directed cycle C in a grid.

Lemma 3.4 (b-value of a cycle). *For any simple directed cycle C in a grid, $b(C) = 0$.*

Proof. We can calculate $b(C) = \sum_{(u,v)\in C} a(u, v)$ alternatively by summing up the b-value of each cell that is inside the cycle, where each cell is seen as a 4-node directed cycle that has the same orientation as that of C, so $b(C) = 0$ by Lemma 3.3. The validity of the calculation is due to the cancellation of the a-values associated with the edges that are strictly inside C, as illustrated in Figure 2. For each directed edge (u, v) on C, $a(u, v)$ appears exactly once in the calculation. For each edge $\{u, v\}$ that is inside C, both $a(u, v)$ and $a(v, u)$ appear exactly once in the calculation, so they cancel each other. □

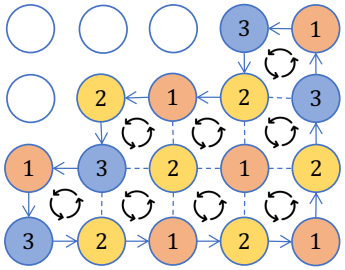


Figure 2: Cancellation of a-values.

For each node $u \in V$, we define

$$i(u) = \begin{cases} 1 & \text{if } c(u) = 3, \\ 0 & \text{otherwise.} \end{cases}$$

as the indicator variable for $c(u) = 3$. We show that the parity of the b-value of a directed path is uniquely determined by the colors of the two endpoints and the parity of the path length.

Lemma 3.5 (Parity of b-value). *Let C be any directed cycle of length ℓ and let $P = (u, \ldots, v)$ be any directed path of length ℓ. We have*

$$b(C) \equiv \ell \pmod 2 \qquad \text{and} \qquad b(P) \equiv i(u) + i(v) + \ell \pmod 2.$$

Proof. We just need to focus on proving $b(P) \equiv i(u) + i(v) + \ell \pmod 2$, as $b(C) \equiv \ell \pmod 2$ follows from $b(P) \equiv i(u) + i(v) + \ell \pmod 2$ by viewing C as a directed path that starts and ends at the same node u and observing that $i(u) + i(u) \pmod 2 = 0$ regardless of the color of u.

Base case. Suppose $\ell = 1$. We claim that $b(P) = a(u, v) \equiv i(u) + i(v) + \ell \pmod 2$. To see this, we divide the analysis into two cases.

- If $c(u) \neq 3$ and $c(v) \neq 3$, then $a(u, v) \in \{-1, 1\}$, so $a(u, v) \equiv i(u) + i(v) + \ell \equiv 1 \pmod 2$, as $i(u) = i(v) = 0$.
- If $c(u) = 3$ or $c(v) = 3$, then $0 = a(u, v) \equiv i(u)+i(v)+\ell \equiv 0 \pmod 2$, as $i(u) + i(v) = 1$.

Summation. Consider the case where $\ell > 1$.

$$\begin{aligned} & b(P) \\ = & \sum_{(s,t)\in P} a(s,t) \\ \equiv & \sum_{(s,t)\in P} (i(s)+i(t)+1) \pmod 2 \quad \text{(the base case)} \\ \equiv & -i(u)-i(v)+2\cdot\sum_{(s,t)\in P}(i(s)+i(t))+\sum_{(s,t)\in P} 1 \pmod 2 \\ \equiv & i(u)+i(v)+\ell \pmod 2. \end{aligned}$$

□

We emphasize that the claim regarding $b(C)$ in lemma 3.5 does *not* follow immediately from lemma 3.4, as lemma 3.4 only considers grids.

Before we continue, we briefly discuss the intuition behind our notion of b-value. This notion intuitively captures the level of difficulty in completing the coloring of the rest of the grid given a colored path. In a proper 3-coloring of a grid, the set of nodes with color 3 separates the remaining nodes into connected regions. In the subsequent discussion, a region refers to a connected set of nodes with colors 1 and 2.

Let us start by considering a directed path $3 \to 2 \to 1 \to 2 \to 1 \to 2 \to 3$ whose b-value is zero. Starting from the coloring of this path, it is easy to extend the partial coloring in a way that the region of nodes colored with 1 and 2 is enclosed by a cycle of nodes colored with 3, see Figure 3.

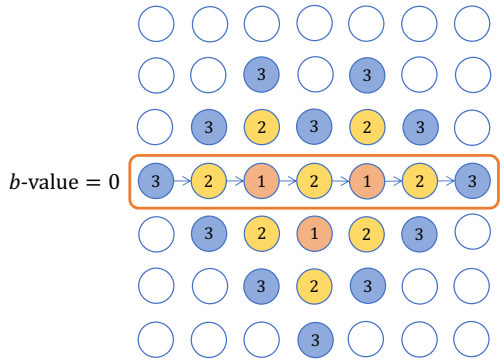


Figure 3: A directed path whose b-value is zero.

Consider a directed path $3 \to 2 \to 1 \to 2 \to 1 \to 3$ whose a b-value is 1. In this case, we cannot close the region with *one* cycle of nodes colored 3. The region of nodes colored with 1 and 2 must continue unless it closes with itself or reaches the boundary of the grid.

Consider any directed cycle C that contains the directed path $3 \to 2 \to 1 \to 2 \to 1 \to 3$ as a subpath. Observe that C must intersect with the region of nodes colored with 1 and 2 *again* via a directed path whose b-value is -1, as illustrated in Figure 4.

Intuitively, the b-value of a directed path counts the difference between the number of occurrences of $3 \to 2 \to \cdots$(colors 1 and 2 appear alternatively)$\cdots \to 1 \to 3$ and the number of occurrences of $3 \to 1 \to \cdots$(colors 1 and 2 appear alternatively)$\cdots \to 2 \to 3$. The above informal discussion suggests that each occurrence $3 \to 2 \to \cdots \to 1 \to 3$ has to be matched with an occurrence of $3 \to 1 \to \cdots \to 2 \to 3$. Therefore, if we can force an algorithm to create a directed path P with a large value of $|b(P)|$, then we may obtain a high locality lower bound for the algorithm.

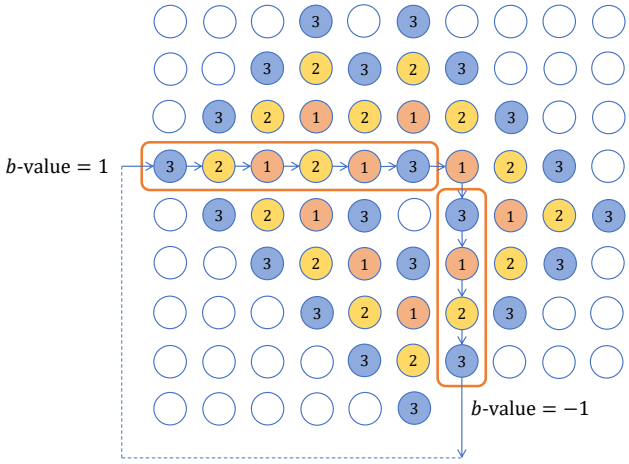


Figure 4: Intersection of a directed cycle and a region of nodes with colors 1 and 2.

3.2 Hardness of 3-Coloring in Simple Grids

In this section, we prove the main result of the paper, i.e., Theorem 1.1. Our proof uses the following properties of b-values in grids.

- The b-value of any directed cycle is *always zero* (Lemma 3.4).
- The parity of the b-value of any directed path is *solely determined* by the parity of the path length and the color of the two endpoints u and v (Lemma 3.5).

To establish Theorem 1.1, we prove that any algorithm designed for 3-coloring grids, operating with a locality of $T(n) \in o(\log n)$, can be strategically countered by an adversary capable of forcing a directed cycle with a non-zero b-value. Throughout this section, let $\mathcal{A}$ be any algorithm for 3-coloring a $(\sqrt{n} \times \sqrt{n})$ grid $G = (V, E)$ with a locality of $T(n) \in o(\log n)$. We analyze the coloring function $c\colon V \to \{1, 2, 3\}$ generated by algorithm $\mathcal{A}$.

The core strategy of our proof is to create a directed path with a substantial b-value, making it impossible for the algorithm to complete the coloring with a small locality. The interaction between the algorithm and the adversary in Online-LOCAL can be seen as a 2-player game as follows.

- The algorithm's task is to label each node v_i based on the current discovered region G_i and the sequence $(v_1, v_2, \ldots, v_i)$. The algorithm wins if the final coloring of G is proper.
- The adversary's task is to select the nodes v_i in the sequence $\sigma = (v_1, v_2, \ldots, v_n)$. Moreover, the adversary has the liberty to adjust how the current discovered region G_i fits into G as an induced subgraph. Informally, suppose G_i consists of several connected components. The adversary has the

flexibility to adjust the *directions* of these components and the *distances* between these components, as the algorithm is unaware of the precise location of these components in G. The adversary wins if the final coloring of G is not proper.

Next, we show that there is an adversary strategy to construct a directed path with a large b-value within a row while keeping the discovered region small. Here we only allow our adversary strategy to select nodes within *one* row, so in the subsequent discussion, we measure the *length* of the current discovered region G_i by the distance between the two farthest nodes in G_i restricted to the row, where the distance is measured with respect to the row.

Lemma 3.6. *Let k be a non-negative integer such that $5^{k+1} \cdot T(n) < \sqrt{n}$. There is an adversary strategy to construct a directed path with its b-value of at least k within a row while keeping the length of the discovered region at most $5^{k+1} \cdot T(n)$.*

Proof. We prove the lemma by an induction on k.

Base case. For the case where $k = 0$, the adversary can just reveal any node v. We view node v as a directed path with a b-value of zero. The length of the discovered region is $2T(n) < 5^{k+1} \cdot T(n)$, as the algorithm has locality $T(n)$.

Inductive step. Consider the case where $k \geq 1$. In the subsequent discussion, we write $P_{x,y}$ to denote the directed path starting from x and ending at y along the row. By the induction hypothesis, we construct two directed paths $P_{u,v}$ and $P_{s,t}$ with a b-value of at least $k-1$. The construction of both directed paths requires a discovered region of length at most $5^k \cdot T(n)$.

We now concatenate these two directed paths together into a directed path $P_{u,t}$ by concatenating the two discovered regions via a path of length $\ell \in \{2, 3\}$ on the row. See Figure 5 for an illustration. After the concatenation, the length of the discovered region is at most $\ell + 2 \cdot 5^k \cdot T(n) \leq 5^{k+1} \cdot T(n)$. Here we use the fact that $T(n) \geq 1$, since 3-coloring cannot be solved with zero locality.

To finish the proof, we just need to select $\ell \in \{2, 3\}$ in such a way that we can find a directed path in the row using the nodes between u and t whose b-value is at least k. If the b-value of one of the paths $P_{u,v}$ and $P_{s,t}$ is already at least k, then we are done, so from now on we assume that their b-values are precisely $k-1$.

By Lemma 3.5, the parity of the b-value of any directed path is *solely determined* by the parity of the path length and the color of the two endpoints. Therefore, we may select $\ell \in \{2, 3\}$ in such a way that the parity of the b-value of the path $P_{v,s}$ differs from $k-1$, i.e., $b(P_{v,s}) \not\equiv k - 1 \pmod 2$.

From now on, we write $h = b(P_{v,s})$, so we have $b(P_{u,t}) = 2(k-1) + h$. We claim that at least one of $|2(k-1)+h|$ or $|h|$ is greater than $|k-1|$. The claim holds true due to the inequality

$$|2(k-1)+h| + |h| = |2(k-1)+h| + |-h| \geq |2(k-1)| = 2 \cdot |k-1|$$

and the observation that the inequality becomes equality *only* when $|h| = |k-1|$, which is impossible because $h \neq k-1$. We conclude that the b-value of at least one of the directed paths $P_{v,s}$, $P_{s,v}$, $P_{u,t}$, and $P_{t,u}$ is at least k, as required. □

We are prepared to prove Theorem 1.1.

Theorem 1.1. *In the* Online-LOCAL *model, the locality of 3-coloring a $(\sqrt{n} \times \sqrt{n})$ grid is $\Omega(\log n)$.*

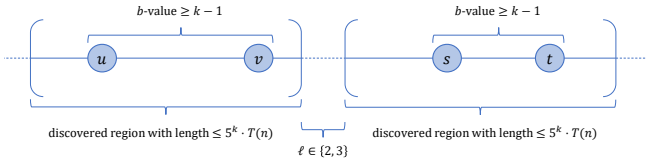


Figure 5: Concatenating $P_{u,v}$ and $P_{s,t}$.

Proof. To prove the theorem, we just need to show that any algorithm $\mathcal{A}$ with locality $T(n) \in o(\log n)$ for 3-coloring a $(\sqrt{n} \times \sqrt{n})$ grid is incorrect. As long as n is at least a sufficiently large constant, we may find an integer k to satisfy the two conditions: $k > 4T(n)+4$ and $5^{k+1} \cdot T(n) < \sqrt{n}$.

In the subsequent discussion, if x and y are two nodes that belong to the same row or the same column, then we write $P_{x,y}$ to denote the directed path starting from x and ending at y along the row or the column.

In the $(\sqrt{n} \times \sqrt{n})$ grid, we apply the adversary strategy of Lemma 3.6 to force the algorithm $\mathcal{A}$ to construct a directed path $P_{u,v}$ in a row R_1 starting from u and ending at v with $b(P_{u,v}) \geq k$. After that, we consider the row R_2 that is above the row R_1 by a distance of $2T(n) + 2$. Let t be the node in R_2 that belongs to the same column as u. Let s be the node in R_2 that belongs to the same column as v. See Figure 6 for an illustration.

As the adversary, we ask the algorithm to color all the nodes in $P_{s,t}$. We may assume that $b(P_{s,t}) \geq 0$. In case $b(P_{s,t}) < 0$, we can simply reverse the direction of $P_{s,t}$. This is possible because the discovered region associated with $P_{s,t}$ is not connected to the discovered region associated with $P_{u,v}$ from the viewpoint of the algorithm, as the distance between R_1 and R_2 is at least $2T(n) + 2$.

We let the algorithm finish the coloring of the remaining nodes arbitrarily. Now, consider the directed cycle C formed by concatenating the four directed paths $P_{u,v}$, $P_{v,s}$, $P_{s,t}$, and $P_{t,u}$. By its definition, the absolute value of the b-value of one path P is at most its length, so the b-value of both $P_{v,s}$ and $P_{t,u}$ is at least $-(2T(n) + 2)$, so

$$\begin{aligned} b(C) &= b(P_{u,v}) + b(P_{v,s}) + b(P_{s,t}) + b(P_{t,u}) \\ &\geq k - (2T(n) + 2) + 0 - (2T(n) + 2) > 0, \end{aligned}$$

which is impossible due to Lemma 3.4, so the coloring produced by the algorithm is incorrect. □

We remark that if we consider a general $(a \times b)$ grid, then the above proof yields an $\Omega(\min\{\log \max\{a, b\}, \min\{a, b\}\})$ locality lower bound. To see this, without loss of generality, assume $b \geq a$, and observe that the proof above works so long as the locality T of the algorithm satisfies $T \in o(\log b)$ and $T \in o(a)$.

3.3 Hardness of 3-Coloring in Toroidal and Cylindrical Grids

In this section, we prove Theorem 1.3.

Theorem 1.3. *In the* Online-LOCAL *model, the locality of 3-coloring $(\sqrt{n} \times \sqrt{n})$ toroidal and cylindrical grids is $\Omega(\sqrt{n})$.*

Proof. We start by discussing the properties of proper 3-coloring of toroidal and cylindrical grids. Consider any two directed

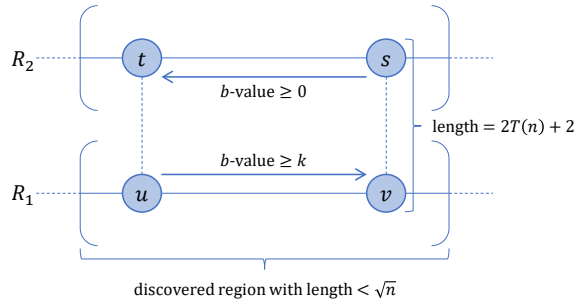


Figure 6: Construction of a directed cycle with non-zero b-value

cycles C_1 and C_2 corresponding to orienting two rows with *different directions*. Same as the proof of Lemma 3.4, we may calculate $b(C_1) + b(C_2)$ alternatively by summing up the b-value for the cells located between C_1 and C_2, so we infer that

$$b(C_1) + b(C_2) = 0. \tag{1}$$

Furthermore, if the number of columns is *odd*, then the lengths of the two directed cycles C_1 and C_2 are odd, so Lemma 3.5 implies that both $b(C_1)$ and $b(C_2)$ are odd numbers.

Consider an arbitrary algorithm $\mathcal{A}$ designed for 3-coloring $(\sqrt{n} \times \sqrt{n})$ cylindrical or toroidal grids with locality $T(n) \in o(\sqrt{n})$. Let us choose a sufficiently large value of n such that $\sqrt{n}$ is an odd number and $4T(n) + 4 \leq \sqrt{n}$. Therefore, we may select two rows such that their $T(n)$-radius neighborhoods induce *non-adjacent* cylindrical grids with $2T(n) + 1$ rows and $\sqrt{n}$ columns.

As the adversary, we instruct algorithm $\mathcal{A}$ to color these two rows. From the perspective of the algorithm, it sees precisely two disjoint cylindrical grids with $2T(n) + 1$ rows and $\sqrt{n}$ columns. Furthermore, the algorithm is unaware of their positions and directions in the input graph. As the adversary, we have the liberty to choose their directions *after* the algorithm fixes the coloring of the two rows. Since the b-value of an odd-length directed cycle is odd, we can always select their directions to violate Equation (1). Hence the algorithm $\mathcal{A}$ cannot correctly 3-color the graph. □

We remark that the $\Omega(\sqrt{n})$ locality lower bound in the proof above comes from the fact that the number of rows is $\sqrt{n}$. If we consider a general $(a \times b)$ toroidal or cylindrical grid, then the above proof yields an $\Omega(a)$ locality lower bound whenever the number of columns b is an odd number.

4 HARDNESS OF $(2k-2)$-COLORING OF k-PARTITE GRAPHS

In this section, we consider the problem of coloring a k-partite graph using $2k-2$ colors and establish a lower bound of $\Omega(n)$ on the locality in the Online-LOCAL model, where $k \geq 2$ is a constant. Thus we prove Theorem 1.5. We introduce a specific type of gadget. Following that, we will proceed with constructing the hard instance graph using these gadgets to establish the lower bound.

Gadget definition. A gadget $A(k)$ is represented as a graph with a set of nodes denoted by $[k] \times [k]$, where each node corresponds to an ordered pair (i, j). Two nodes, (i, j) and (i', j'), are connected by an edge if and only if $i \neq i'$ and $j \neq j'$. Furthermore, for any $i \in [k]$, the set of nodes $\{(i, j) \colon j \in [k]\}$ defines the i-th row of the gadget $A(k)$. Similarly, for $j \in [k]$, the set of nodes $\{(i, j) \colon i \in [k]\}$ defines the j-th column of the gadget. Therefore, an edge exists between two nodes in a gadget if and only if they are neither in the same row nor the same column.

Construction of the hard instance graph G^*. G^* comprises of $n' = n/k^2$ gadgets denoted as $A_1, \ldots, A_{n'}$, where the node set of the ℓ-th gadget A_ℓ can be represented as $\{\ell\} \times [k] \times [k]$. Consequently, the node set of G^* is $[n'] \times [k] \times [k]$. The edge set of G^* is defined as follows.

- **Edges within the gadgets:** For each $\ell \in [n']$, there is an edge between (ℓ, i, j) and (ℓ, i', j') if and only if $i \neq i'$ and $j \neq j'$.
- **Edges between consecutive gadgets:** For each $\ell \in [n' - 1]$, there is an edge between (ℓ, i, j) and $(\ell + 1, i', j')$ if and only if $i \neq i'$ and $j \neq j'$.

Proposition 4.1. *The graph G^* is k-partite.*

Proof. We will demonstrate a proper coloring of G^* with a set of k colors $\{1, \ldots, k\}$. Recall the construction of G^* that consists of gadgets $A_1, \ldots, A_{n'}$. For any gadget, color all the nodes in the i-th row using color i, where $i \in [k]$. This coloring is proper due to the following reason: An edge between two nodes, either in the same gadget or in consecutive gadgets, can only exist if they are neither in the same row nor the same column. □

Theorem 1.5. *Let $k \geq 2$ be a constant. The locality of $(2k-2)$-coloring a k-partite graph is $\Omega(n)$ in the* Online-LOCAL *model.*

Proof. Due to the page limit, we only sketch the high-level idea behind the proof. See the full version [9] of the paper for a complete proof. Given a proper coloring of a gadget, it is said to be *row-colorful* if it has a row where all nodes in the row are colored with distinct colors. Similarly, we can define a *column-colorful* gadget. We show that, for a proper coloring of a gadget with $2k-2$ colors, the gadget is precisely one out of row-colorful and column-colorful. Furthermore, leveraging the above properties of the gadgets, we establish a crucial property of the graph G^* with gadgets $A_1, \ldots, A_{n'}$: For any proper $(2k-2)$-coloring of G^*, all the gadgets are either row-colorful or all of them are column-colorful. To establish the lower bound, consider a scenario where the adversary asks the algorithm to first color the nodes in A_1, followed by the nodes in $A_{n'}$, and then the remaining nodes. Regardless of how the algorithm, in the Online-LOCAL model with locality $o(n)$, colors the nodes in A_1 and $A_{n'}$, we argue that the adversary can modify the input graph in a specific way. In particular, the modification results in an input graph that is isomorphic to G^*, consistent with the previously explored part of the graph, and crucially ensures that A_1 is row-colorful while $A_{n'}$ is column-colorful. However, this leads to a failure to color some nodes in the future, as either all the gadgets in the input graph are row-colorful or all of them are column-colorful. □

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